

Final Project Report

Submitted By

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Internship Program

Bluestock Fintech

Technologies Used

Python | Pandas | NumPy | SQL | SQLite | Power BI | Matplotlib

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Executive Summary

The Indian mutual fund industry has emerged as one of the fastest-growing segments within the financial services sector. Increased investor awareness, digital investment platforms, regulatory transparency, and the widespread adoption of Systematic Investment Plans (SIPs) have significantly accelerated industry growth. As a result, mutual fund organizations now generate vast quantities of data related to fund performance, investor participation, portfolio composition, and market benchmarks. Converting this information into meaningful insights requires robust analytical frameworks capable of integrating data engineering, financial analytics, and business intelligence.

The Bluestock Mutual Fund Analytics Capstone was developed as a comprehensive end-to-end analytics solution designed to address these requirements. The project integrates multiple datasets covering mutual fund schemes, historical Net Asset Values (NAVs), Assets Under Management (AUM), SIP inflows, investor transactions, portfolio holdings, and benchmark indices. Through a structured Extract-Transform-Load (ETL) process, raw datasets were converted into analytics-ready structures suitable for reporting and advanced financial analysis.

The analytical environment successfully processed more than 46,000 NAV observations, 32,778 investor transactions, and 40 mutual fund schemes across multiple categories. The project implemented industry-standard performance metrics including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These metrics were used to evaluate mutual fund quality from both return and risk perspectives.

Beyond traditional performance evaluation, advanced risk analytics were developed using Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio analysis, Investor Cohort Analysis, SIP Continuity Analysis, and Portfolio Concentration Analysis using the Herfindahl-Hirschman Index (HHI). A rule-based recommendation engine was also implemented to assist investors in identifying suitable funds based on risk preferences.

The final deliverables include analytical reports, financial risk models, an investor recommendation engine, and a four-page Power BI dashboard providing interactive business intelligence capabilities. The resulting platform enables users to evaluate fund performance, monitor industry growth, understand investor behavior, and make more informed investment decisions.

1. Introduction

1.1 Industry Background

Mutual funds play a critical role in modern investment ecosystems by providing diversified exposure to financial markets through professionally managed portfolios. They allow investors to participate in equity, debt, hybrid, and sector-specific investment opportunities without requiring extensive market expertise. Over the last decade, India has witnessed significant growth in mutual fund participation due to rising financial literacy, increased accessibility through digital platforms, and favorable regulatory initiatives.

The expansion of the mutual fund industry has resulted in exponential growth in Assets Under Management (AUM), SIP contributions, investor accounts, and fund offerings. While this growth creates new opportunities, it also introduces analytical challenges. Investors must evaluate funds not only on the basis of returns but also on risk-adjusted performance, downside exposure, concentration risk, and consistency of performance over time.

Traditional investment evaluation methods often emphasize returns while overlooking critical risk factors. Modern investment decision-making requires a more holistic framework that incorporates quantitative analytics, statistical risk measures, and business intelligence reporting. The growing availability of financial data provides an opportunity to build such frameworks and generate meaningful insights for both investors and financial institutions.

The Bluestock Mutual Fund Analytics Capstone addresses this requirement by integrating data engineering, financial analytics, and visualization techniques into a unified analytical platform.

1.2 Project Overview

The objective of this project was to design and develop a complete mutual fund analytics ecosystem capable of processing, analyzing, and visualizing financial datasets from multiple sources. The project was executed through a structured seven-day implementation plan covering data collection, ETL development, exploratory analysis, performance analytics, advanced risk modeling, dashboard creation, and final reporting.

The solution combines Python-based analytics with Power BI visualization to provide users with an intuitive and comprehensive understanding of mutual fund performance and investor behavior. By integrating multiple datasets into a centralized environment, the project enables business users to perform comparative analysis, identify trends, evaluate risk exposure, and generate actionable insights.

The final solution is designed to support multiple stakeholders, including retail investors, analysts, portfolio managers, and business decision-makers.

2. Problem Statement

The mutual fund ecosystem generates large amounts of fragmented data originating from independent operational systems. Information relating to fund performance, investor transactions, portfolio composition, benchmark indices, and industry inflows often exists in separate formats and structures. This fragmentation limits the ability of investors and analysts to derive comprehensive insights and make data-driven decisions.

Several challenges arise from this situation:

- Difficulty comparing mutual fund schemes using standardized performance metrics.
- Limited visibility into risk-adjusted performance and downside risk.
- Lack of centralized reporting for investor behavior analysis.
- Insufficient understanding of SIP growth and retention trends.
- Difficulty identifying portfolio concentration risk.
- Absence of integrated business intelligence tools for monitoring industry performance.

Without a unified analytical framework, valuable insights remain hidden within raw datasets, reducing the effectiveness of investment analysis and strategic decision-making.

This project seeks to address these challenges through the development of a centralized analytics platform capable of integrating multiple datasets, calculating advanced

performance metrics, performing risk analysis, and delivering business-friendly visualizations.

3. Project Objectives

The primary objectives of the Bluestock Mutual Fund Analytics Capstone are divided into five categories.

Data Engineering Objectives

- Develop a robust ETL pipeline for processing mutual fund datasets.
- Validate and clean raw data sources.
- Standardize and transform data into analytics-ready formats.
- Store processed information within a centralized SQLite database.

Exploratory Data Analysis Objectives

- Analyze industry growth trends.
- Evaluate Assets Under Management (AUM) growth patterns.
- Study SIP contribution trends.
- Examine investor demographics and transaction behavior.
- Understand category-wise investment flows.

Performance Analytics Objectives

- Calculate Compound Annual Growth Rate (CAGR).
- Measure Sharpe Ratio and Sortino Ratio.
- Calculate Alpha and Beta relative to benchmark performance.
- Measure Maximum Drawdown.
- Develop fund scorecards for comparative evaluation.

Advanced Analytics Objectives

- Calculate Value at Risk (VaR).
- Calculate Conditional Value at Risk (CVaR).
- Analyze Rolling Sharpe Ratios.

- Conduct Investor Cohort Analysis.
- Perform SIP Continuity Analysis.
- Measure Portfolio Concentration Risk using HHI.

Business Intelligence Objectives

- Build interactive Power BI dashboards.
 - Generate actionable business insights.
 - Support investment decision-making through data visualization.
 - Deliver professional analytical reports and presentations.
-

4. Dataset Overview

The project integrates ten datasets representing multiple dimensions of mutual fund analytics. These datasets collectively provide information relating to fund characteristics, performance, investor activity, portfolio composition, and industry trends.

The primary datasets used in the project include:

Dataset	Purpose
Fund Master	Scheme metadata and classification
NAV History	Historical fund performance
AUM Data	Industry growth analysis
SIP Inflows	Monthly SIP contribution tracking
Category Inflows	Category-wise investment analysis
Investor Transactions	Investor behavior analysis
Portfolio Holdings	Concentration risk analysis
Benchmark Indices	Comparative performance analysis
Folio Counts	Investor participation analysis
Performance Metrics	Risk-adjusted evaluation

The analytical environment processed:

- 40 mutual fund schemes
- More than 46,000 NAV observations
- 32,778 investor transactions
- 8,050 benchmark observations
- Multiple years of historical investment data

These datasets provide a comprehensive view of mutual fund performance, investor participation, industry growth, and portfolio risk.

Figure 1 Placement

Insert:

Project Architecture Diagram

(Center of Page)

Caption:

Figure 1: End-to-End Mutual Fund Analytics Architecture

This diagram should illustrate:

Raw Data Sources → ETL Pipeline → SQLite Database → Analytics Engine → Power BI Dashboard → Business Insights

5. System Architecture

The Bluestock Mutual Fund Analytics platform was designed using a layered architecture to ensure scalability, maintainability, and analytical efficiency. The architecture separates data collection, processing, storage, analytics, and visualization into distinct components, enabling a structured flow of information from raw datasets to business insights.

The system begins with multiple raw data sources containing mutual fund information, NAV history, investor transactions, SIP inflows, portfolio holdings, benchmark indices, and industry statistics. These datasets are ingested through Python-based ETL pipelines that perform cleaning, transformation, and validation operations.

After processing, the datasets are loaded into a centralized SQLite database that acts as the analytical repository for downstream operations. The database serves as the single source of truth and ensures data consistency across analytical workflows.

The analytics layer utilizes Python libraries including Pandas, NumPy, and Matplotlib to perform exploratory data analysis, performance analytics, and advanced risk modeling. Finally, Power BI dashboards consume the processed datasets and analytical outputs to provide business users with interactive reporting capabilities.

Figure 2 Placement

Insert Image:

Architecture_Diagram.png

Caption:

Figure 2: End-to-End Mutual Fund Analytics Architecture

The architecture diagram should visually represent the following flow:

Raw Data Sources → ETL Pipeline → SQLite Database → Analytics Engine → Power BI Dashboard → Business Insights

6. ETL Pipeline

6.1 Overview

Data engineering forms the foundation of any successful analytics project. Before meaningful insights can be generated, raw datasets must be cleaned, standardized, validated, and transformed into structured formats suitable for analysis.

The ETL (Extract, Transform, Load) process developed for this project ensures that all datasets are processed consistently and meet quality standards before analytical operations are performed.

The ETL pipeline was implemented using Python and consists of three major stages:

1. Data Extraction
2. Data Transformation
3. Data Loading

This structured workflow minimizes data inconsistencies and improves analytical reliability.

6.2 Data Extraction

The extraction stage focused on collecting information from multiple independent datasets covering different aspects of the mutual fund ecosystem.

The datasets included:

- Fund Master Information
- NAV History
- AUM Records
- SIP Inflow Data
- Category Inflows
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices
- Folio Counts

Each dataset was imported into Python using Pandas dataframes and inspected for structural consistency.

The extraction phase successfully integrated ten independent datasets and created a unified analytical environment.

6.3 Data Transformation

The transformation stage focused on improving data quality and ensuring compatibility across datasets.

Several transformation operations were performed:

Missing Value Handling

Missing values were identified and treated using appropriate strategies based on the nature of the dataset.

Methods included:

- Null value replacement
- Record filtering
- Default value assignment

This ensured that incomplete records did not negatively impact analytical results.

Duplicate Detection

Duplicate records were identified and removed to prevent distortions in reporting and metric calculations.

Examples included:

- Duplicate transaction records
- Repeated NAV observations
- Redundant portfolio entries

The deduplication process improved data accuracy and consistency.

Data Type Standardization

Many datasets contained inconsistencies in data formats.

The following conversions were performed:

- Date columns converted to datetime format
- Numerical fields converted to numeric data types
- Percentage fields standardized
- Text fields normalized

Standardized data types enabled efficient analytical processing and database storage.

Column Normalization

Several datasets contained inconsistent naming conventions.

Column names were standardized to:

- Improve readability
- Simplify joins
- Maintain consistency across datasets

This step significantly improved maintainability of analytical scripts.

Validation Rules

Business validation rules were implemented to ensure logical consistency.

Examples included:

- Valid AMFI codes
- Positive NAV values
- Non-negative SIP amounts
- Valid transaction dates

Records violating validation rules were flagged for further review.

6.4 Data Loading

Following transformation and validation, the cleaned datasets were loaded into SQLite.

SQLite was selected because:

- Lightweight architecture
- Easy integration with Python
- Efficient query execution
- Suitable for analytical workloads

The database acts as a centralized repository and enables consistent access across notebooks, scripts, and dashboard components.

7. Data Quality Assessment

Data quality is critical for producing reliable analytical outputs. Poor-quality data can lead to incorrect conclusions and reduce confidence in business decisions.

A comprehensive quality assessment was conducted during the ETL phase.

The assessment focused on:

- Completeness
 - Consistency
 - Accuracy
 - Uniqueness
 - Validity
-

7.1 Completeness Analysis

Completeness measures the extent to which required data fields contain valid values.

The assessment revealed:

- High completeness across fund-level datasets.
- Minimal missing values in transaction records.
- Strong coverage of NAV history data.

The overall completeness level was sufficient for advanced analytics.

7.2 Consistency Analysis

Consistency checks ensured that values remained logically compatible across datasets.

Examples included:

- Matching AMFI codes across tables.
- Consistent date formats.
- Valid category mappings.
- Correct benchmark references.

The consistency assessment identified no major structural issues.

7.3 Accuracy Analysis

Accuracy checks were performed through:

- AMFI code validation
- Range validation
- Duplicate detection
- Cross-table verification

These procedures increased confidence in analytical outputs.

7.4 Data Quality Summary

The ETL process successfully produced analytics-ready datasets with:

- Minimal missing values
- No major duplicate issues
- Standardized formats
- Validated identifiers
- High structural consistency

As a result, the processed datasets were suitable for performance analytics, risk modeling, and dashboard reporting.

8. Database Design

A centralized SQLite database was developed to serve as the analytical backbone of the project.

The database stores processed datasets and supports querying, analysis, and dashboard integration.

The schema was designed to maintain relationships between mutual fund entities and transactional records.

Major tables include:

Fund Master

Stores scheme metadata including:

- AMFI Code
 - Scheme Name
 - Category
 - Fund House
-

NAV History

Stores historical NAV observations for each scheme.

Key fields:

- Date
 - NAV
 - AMFI Code
-

Transactions

Stores investor transaction records.

Key fields:

- Investor ID
 - Transaction Date
 - Amount
 - Transaction Type
-

SIP Inflows

Stores monthly SIP contribution data.

Key fields:

- Month
 - SIP Amount
-

Portfolio Holdings

Stores sector allocation information.

Used for HHI concentration analysis.

Benchmark Data

Stores benchmark index observations.

Used for:

- Alpha calculations
 - Beta calculations
 - Relative performance analysis
-

9. Validation Results

After completing the ETL process and database loading operations, a final validation stage was performed.

Validation confirmed:

- Successful integration of all ten datasets.
- Correct AMFI code mappings.
- Consistent date structures.
- Valid numerical fields.
- Accurate database loading.

The final analytical environment contained:

- 40 mutual fund schemes
- More than 46,000 NAV observations
- 32,778 investor transactions
- 8,050 benchmark observations

These validated datasets formed the foundation for all subsequent analytical activities including EDA, performance evaluation, advanced risk analytics, and dashboard reporting.

Figure 3 Placement

Insert Screenshot:

Database_Verification_Output.png (if available)

Caption:

Figure 3: Database Validation and Record Verification Results

10. Exploratory Data Analysis (EDA)

10.1 Overview

Exploratory Data Analysis (EDA) was conducted to understand the structure, characteristics, and underlying patterns within the mutual fund datasets. The objective of EDA was not only to summarize historical information but also to identify meaningful trends, investor behavior patterns, and performance characteristics that could support later stages of analysis.

The EDA phase focused on answering several business questions:

- How has the mutual fund industry grown over time?
- Which fund houses dominate industry assets?
- How are SIP contributions changing?
- Which investor segments contribute the most capital?
- How do transaction patterns vary across demographics and geography?

The insights generated during this phase formed the foundation for both performance analytics and dashboard development.

11. Industry Overview Analysis

Figure 4 Placement

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Industry_Overview.png

Caption:

Figure 4: Industry Overview Dashboard

11.1 Industry Growth Overview

The Industry Overview dashboard was designed to provide a high-level summary of the mutual fund ecosystem and industry performance.

The dashboard consolidated multiple KPIs into a single view, allowing users to quickly assess industry scale and participation levels.

The key metrics included:

- Total Assets Under Management (AUM)
- Total SIP Contributions
- Total Folio Count
- Number of Mutual Fund Schemes

The dashboard revealed that the industry had achieved substantial scale, reflecting increasing investor participation and broader market adoption of mutual fund products.

11.2 Assets Under Management (AUM) Analysis

Assets Under Management represent one of the most important indicators of industry growth and investor confidence.

The analysis showed that AUM expanded consistently across the observation period, indicating:

- Increasing investor participation
- Growth in long-term investments
- Expansion of mutual fund penetration

Several major fund houses emerged as dominant players within the industry.

Key Observations

- SBI Mutual Fund maintained one of the highest AUM values.
- Leading fund houses collectively accounted for a significant share of industry assets.
- Industry-wide AUM growth remained positive despite periodic market fluctuations.

Business Interpretation

Higher AUM levels generally indicate stronger investor confidence and greater institutional credibility. The concentration of assets among leading fund houses reflects investor preference for established brands and long-term performance records.

11.3 Fund House Analysis

The dashboard provided a comparative view of major fund houses and their relative market positions.

Analysis revealed clear differences in market share across asset management companies (AMCs).

Findings

- A small number of AMCs controlled a large proportion of total industry assets.
- Established fund houses maintained a competitive advantage due to investor trust and product diversity.

- Mid-sized AMCs demonstrated growth potential through specialized fund offerings.

Business Interpretation

The concentration of industry assets among leading AMCs suggests that brand reputation and historical performance remain important factors influencing investor decisions.

12. SIP Trend Analysis

Figure 5 Placement

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SIP_Market_Trends.png

Caption:

Figure 5: SIP and Market Trends Dashboard

12.1 Importance of SIP Analysis

Systematic Investment Plans (SIPs) have become one of the most popular investment mechanisms within the mutual fund industry. SIPs encourage disciplined investing and allow investors to participate in financial markets through regular contributions.

Analyzing SIP trends provides valuable insights into:

- Investor confidence
 - Retail participation
 - Long-term investment behavior
 - Market sentiment
-

12.2 Monthly SIP Growth

The dashboard revealed a consistent increase in SIP contributions across the analysis period.

Key Findings

- Monthly SIP inflows demonstrated a clear upward trend.
- Investor participation expanded steadily.

- Periodic fluctuations were observed but did not alter the long-term growth trajectory.

Business Interpretation

The sustained increase in SIP contributions indicates growing financial awareness and increasing adoption of long-term investment strategies among retail investors.

This trend also highlights the resilience of SIP-based investing, which continues to attract capital even during periods of market volatility.

12.3 Category-Wise Inflows

Category-level analysis revealed differences in investor preferences across mutual fund categories.

Findings

- Certain categories attracted significantly larger net inflows than others.
- Liquid and diversified investment categories remained popular among investors.
- Sector-specific categories exhibited greater variability.

Business Interpretation

Investor preference for diversified and liquid investment options suggests a focus on balancing returns with risk management.

12.4 Benchmark Comparison

Benchmark analysis was incorporated into the dashboard to evaluate mutual fund performance relative to broader market movements.

Findings

- Several funds closely tracked benchmark performance.
- Performance divergence was observed among actively managed schemes.
- Benchmark comparisons improved the interpretability of fund returns.

Business Interpretation

Benchmark analysis enables investors to distinguish between market-driven performance and manager-driven performance, improving investment evaluation.

13. Investor Analytics

Figure 6 Placement

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Investor_Analytics.png

Caption:

Figure 6: Investor Analytics Dashboard

13.1 Overview

Investor behavior analysis is essential for understanding participation patterns, demographic characteristics, and transaction preferences.

The Investor Analytics dashboard was developed to provide insights into:

- Age distribution
 - Transaction behavior
 - Geographic participation
 - Investment preferences
-

13.2 Age Group Analysis

The dashboard identified significant differences in investment activity across age groups.

Findings

The 26–35 age group emerged as the most active investor segment.

This demographic accounted for the highest transaction volume and contributed a substantial proportion of total investment activity.

Business Interpretation

The strong participation of younger investors can be attributed to:

- Increasing financial literacy
- Greater adoption of digital investment platforms

- Longer investment horizons
- Higher willingness to participate in equity-oriented products

Strategic Implication

Fund houses should prioritize products and marketing campaigns targeted toward younger investors, as they represent the largest growth opportunity within the industry.

13.3 Transaction Type Analysis

The dashboard categorized transactions into major investment activities including:

- SIP Investments
- Lump Sum Investments
- Redemptions

Findings

- Lump Sum transactions contributed the largest share of transaction value.
- SIP transactions represented a significant source of recurring investment activity.
- Redemption activity remained comparatively lower.

Business Interpretation

The coexistence of large lump-sum investments and recurring SIP contributions indicates a balanced investor base consisting of both long-term investors and tactical market participants.

13.4 Geographic Distribution

State-level analysis provided insights into regional participation patterns.

Findings

High levels of investment activity were observed in:

- Punjab
- Tamil Nadu
- Gujarat

- Rajasthan
- Delhi

Business Interpretation

These states represent important markets for investor acquisition and financial product distribution.

Regional analysis can help AMCs design localized marketing strategies and improve investor engagement.

14. Key EDA Insights

The exploratory analysis phase generated several important findings:

Insight 1

The mutual fund industry demonstrated strong growth in both assets and investor participation.

Insight 2

SIP contributions increased steadily, reflecting growing adoption of disciplined investing practices.

Insight 3

Industry assets remained concentrated among a small number of major fund houses.

Insight 4

The 26–35 age group represented the most active investor segment.

Insight 5

Geographic participation patterns revealed significant opportunities for targeted investor outreach.

Insight 6

Benchmark comparisons highlighted meaningful differences in fund management effectiveness.

Transition to Performance Analytics

The insights generated during the exploratory analysis phase established a strong foundation for the next stage of the project: Performance Analytics.

While EDA focused on understanding industry trends and investor behavior, the performance analytics phase evaluates individual mutual fund schemes using quantitative measures such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.

These metrics provide a deeper understanding of fund quality, consistency, and risk-adjusted performance.

15. Performance Analytics

15.1 Overview

While exploratory analysis focuses on understanding industry trends and investor behavior, performance analytics evaluates the effectiveness of individual mutual fund schemes. Investors require objective measures that assess not only returns but also the level of risk undertaken to generate those returns.

To address this requirement, multiple industry-standard performance metrics were implemented within the project. These metrics provide a comprehensive view of mutual fund quality and allow meaningful comparisons across schemes.

The key metrics calculated include:

- Compound Annual Growth Rate (CAGR)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Together, these measures provide insights into growth potential, risk-adjusted returns, benchmark outperformance, volatility, and downside exposure.

16. Compound Annual Growth Rate (CAGR) Analysis

16.1 Introduction

CAGR is one of the most widely used metrics for evaluating investment performance over multiple years. Unlike simple average returns, CAGR accounts for compounding effects and provides a more realistic representation of long-term growth.

The formula used is:

$$\text{CAGR} = (\text{Ending Value} / \text{Beginning Value})^{(1/n)} - 1$$

where n represents the number of years.

16.2 Findings

The project calculated CAGR for all mutual fund schemes included in the analysis.

Results revealed substantial variation in long-term growth performance across schemes.

Several funds demonstrated strong annualized growth rates exceeding industry averages, indicating effective fund management and favorable market positioning.

The highest-performing schemes achieved CAGR values exceeding 30%, reflecting strong wealth creation potential over the observation period.

16.3 Business Interpretation

A higher CAGR indicates stronger long-term performance and more effective capital appreciation.

However, CAGR alone does not account for volatility or downside risk. Therefore, additional risk-adjusted metrics were required to evaluate overall fund quality.

Figure 7 Placement

Insert:

top_funds_return.png

Caption:

Figure 7: Top Mutual Funds Ranked by Long-Term Performance

17. Sharpe Ratio Analysis

17.1 Introduction

The Sharpe Ratio measures the amount of excess return generated for each unit of risk undertaken.

It is calculated as:

Sharpe Ratio = (Portfolio Return – Risk Free Rate) / Portfolio Standard Deviation

The Sharpe Ratio is widely regarded as one of the most important measures of risk-adjusted performance.

17.2 Findings

The analysis identified meaningful differences in risk-adjusted returns across mutual fund schemes.

Several funds consistently achieved Sharpe Ratios above 1.0, indicating strong performance relative to volatility.

The highest Sharpe Ratios observed in the project ranged between approximately 1.31 and 1.45.

Funds with higher Sharpe Ratios demonstrated the ability to generate superior returns without proportionally increasing portfolio risk.

17.3 Business Interpretation

A high Sharpe Ratio indicates that a fund is delivering attractive returns relative to the amount of volatility experienced.

For investors, this metric provides a more balanced assessment of performance than return-based measures alone.

Funds with strong Sharpe Ratios are generally more attractive for long-term investment because they demonstrate efficient risk utilization.

18. Sortino Ratio Analysis

18.1 Introduction

While the Sharpe Ratio considers total volatility, the Sortino Ratio focuses specifically on downside volatility.

This distinction is important because investors are generally more concerned about losses than positive fluctuations.

The Sortino Ratio is calculated using only negative returns in the denominator.

18.2 Findings

The analysis revealed that some funds achieved significantly stronger Sortino Ratios than their Sharpe Ratios would suggest.

This indicates that negative volatility was relatively limited despite overall fluctuations.

Several top-performing funds demonstrated strong downside-risk-adjusted performance, making them attractive candidates for risk-conscious investors.

18.3 Business Interpretation

The Sortino Ratio provides a more investor-centric evaluation of performance by focusing exclusively on harmful volatility.

Funds with strong Sortino Ratios may be preferable for investors seeking stable long-term growth while minimizing downside exposure.

19. Alpha and Beta Analysis

19.1 Introduction

Alpha and Beta are widely used to evaluate mutual fund performance relative to benchmark indices.

Alpha

Alpha measures the excess return generated by a fund relative to what would be expected based on market performance.

Positive Alpha indicates outperformance.

Beta

Beta measures sensitivity to market movements.

A Beta greater than 1 indicates higher volatility than the market, while a Beta below 1 indicates lower volatility.

19.2 Findings

The analysis identified meaningful differences in benchmark-relative performance across schemes.

Several funds generated positive Alpha values, suggesting superior fund management and stock selection capabilities.

Beta values varied significantly, indicating different levels of market sensitivity across fund categories.

19.3 Business Interpretation

Funds with positive Alpha demonstrate an ability to outperform benchmarks through active management.

Meanwhile, Beta provides investors with an indication of how aggressively a fund responds to market movements.

Together, these metrics support better portfolio construction and risk management decisions.

20. Maximum Drawdown Analysis

20.1 Introduction

Maximum Drawdown measures the largest peak-to-trough decline experienced by an investment.

Unlike volatility metrics, drawdown focuses on actual losses that investors would have experienced.

It is one of the most important indicators of downside risk.

20.2 Findings

The analysis revealed substantial differences in downside exposure across mutual fund schemes.

Several funds maintained relatively low drawdown values despite strong return performance.

The strongest-performing funds experienced drawdowns as low as approximately -12.97%, indicating effective risk management.

Other funds exhibited significantly deeper declines, highlighting greater vulnerability during adverse market conditions.

20.3 Business Interpretation

Investors often evaluate investments based on potential gains while underestimating downside risk.

Maximum Drawdown provides a practical measure of loss potential and helps investors select funds aligned with their risk tolerance.

Funds with lower drawdowns are generally more suitable for conservative investors.

21. Fund Scorecard Development

21.1 Methodology

To provide a consolidated view of fund quality, a composite scorecard was developed.

The scorecard incorporated:

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Each metric was normalized and combined into an overall ranking system.

This approach allowed funds to be evaluated using multiple performance dimensions simultaneously.

21.2 Top Ranked Funds

The scorecard identified the strongest mutual fund schemes based on overall performance and risk characteristics.

Top Fund Rankings

AMFI Code Composite Score

120505 77.75

148569 75.25

119598 75.13

149324	74.75
120843	71.88

21.3 Interpretation

The top-ranked funds consistently demonstrated:

- Strong long-term growth
- Attractive risk-adjusted returns
- Positive benchmark-relative performance
- Controlled downside exposure

These characteristics make them suitable candidates for long-term portfolio inclusion.

Figure 8 Placement

Insert:

fund_scorecard_chart.png (or equivalent ranking chart)

Caption:

Figure 8: Top Ranked Mutual Funds Based on Composite Scorecard

Transition to Advanced Analytics

While performance analytics provides valuable insights into returns and risk-adjusted performance, additional techniques are required to understand downside risk, investor behavior, portfolio concentration, and market resilience.

The next phase of the project focuses on advanced analytics, including:

- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Portfolio Concentration Risk (HHI)

- Fund Recommendation Engine

These analytical models provide a deeper understanding of risk and investor dynamics beyond traditional performance metrics.

22. Advanced Analytics

22.1 Overview

Traditional performance metrics such as CAGR and Sharpe Ratio provide valuable insights into fund quality. However, modern investment analysis requires a deeper understanding of downside risk, investor behavior, portfolio concentration, and performance consistency.

To address these requirements, several advanced analytical models were developed during the project:

- Historical Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Portfolio Concentration Analysis (HHI)
- Fund Recommendation Engine

These techniques extend beyond conventional performance evaluation and provide a more comprehensive view of investment risk and investor participation patterns.

23. Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR)

23.1 Introduction

Value at Risk (VaR) is one of the most widely used risk metrics in financial analysis. It estimates the maximum expected loss that may occur within a specified confidence interval under normal market conditions.

For this project, Historical VaR was calculated using the 5th percentile of daily return distributions.

Conditional Value at Risk (CVaR), also known as Expected Shortfall, measures the average loss that occurs when returns fall below the VaR threshold.

Together, these metrics provide a realistic assessment of downside risk exposure.

23.2 Methodology

Historical VaR (95%) was calculated as:

VaR = 5th Percentile of Daily Returns

CVaR was calculated as:

CVaR = Mean(Returns Below VaR)

The analysis was performed for all mutual fund schemes included in the project.

23.3 Findings

The analysis revealed meaningful differences in downside risk across schemes.

Highest Risk Funds

AMFI Code VaR (95%)

119599	-2.69%
119095	-2.62%
101207	-2.60%
118634	-2.54%
119598	-2.45%

These funds demonstrated the greatest downside exposure under adverse market conditions.

Business Interpretation

Funds with larger negative VaR values may generate attractive returns during favorable market periods but also expose investors to greater short-term losses.

Risk-sensitive investors should carefully evaluate such schemes before investing.

Figure 9 Placement

Insert:

var_cvar_chart.png

Caption:

Figure 9: Historical VaR and CVaR Comparison Across Mutual Funds

24. Rolling Sharpe Ratio Analysis

24.1 Purpose

While traditional Sharpe Ratios provide a single performance value, they do not reveal how performance evolves through time.

Rolling Sharpe analysis addresses this limitation by calculating Sharpe Ratios continuously across moving windows.

A 90-day rolling window was used within this project.

24.2 Findings

The analysis revealed significant variation in risk-adjusted performance over time.

Some funds maintained consistently positive Sharpe Ratios while others experienced substantial fluctuations.

Market volatility had a noticeable impact on risk-adjusted returns, particularly during periods of increased uncertainty.

Business Interpretation

Rolling Sharpe analysis provides investors with insights into the stability and consistency of performance.

Funds with stable Sharpe Ratios are generally considered more reliable investment options.

Figure 10 Placement

Insert:

rolling_sharpe_chart.png

Caption:

Figure 10: Rolling 90-Day Sharpe Ratio Analysis

25. Investor Cohort Analysis

25.1 Objective

Investor Cohort Analysis groups investors based on their first investment year and evaluates behavioral patterns across different cohorts.

The objective is to understand how participation, investment volume, and SIP activity evolve over time.

25.2 Findings

Two major cohorts emerged from the analysis.

2024 Cohort

- Average SIP Amount: ₹107,422
- Total Invested: ₹3.49 Billion

2025 Cohort

- Average SIP Amount: ₹109,159
 - Total Invested: ₹30.45 Million
-

25.3 Interpretation

Although average SIP amounts were comparable across cohorts, investors who entered during 2024 contributed substantially more capital overall.

This difference can largely be attributed to longer participation periods and a larger investor base.

Business Implication

Investor retention strategies should focus on maintaining long-term engagement because earlier cohorts generate significantly greater lifetime value.

Figure 11 Placement

Insert:

cohort_analysis_chart.png

Caption:

Figure 11: Investor Cohort Analysis

26. SIP Continuity Analysis

26.1 Objective

Consistent SIP participation is an important indicator of investor retention and long-term engagement.

The objective of SIP Continuity Analysis was to identify investors at risk of discontinuing their SIP contributions.

26.2 Methodology

The analysis focused on investors with six or more SIP transactions.

The average gap between consecutive SIP contributions was calculated.

Investors with average gaps greater than 35 days were classified as:

"At Risk"

26.3 Findings

The analysis successfully identified investor groups exhibiting irregular contribution patterns.

These investors may require targeted engagement initiatives to prevent SIP discontinuation.

Business Implication

Proactive retention campaigns and automated reminders can significantly improve SIP continuity rates and long-term investor participation.

Figure 12 Placement

Insert:

sip_continuity_chart.png

Caption:

Figure 12: SIP Continuity and Investor Retention Analysis

27. Portfolio Concentration Analysis (HHI)

27.1 Introduction

Portfolio diversification plays a critical role in risk management.

To evaluate diversification levels, the Herfindahl-Hirschman Index (HHI) was calculated using sector allocation weights.

The formula is:

$HHI = \sum(weight^2)$

Higher values indicate greater concentration risk.

27.2 Findings

Several funds exhibited elevated concentration levels.

Highest HHI Funds

AMFI Code HHI

119092 2064

101207 2007

119599 1747

102885 1747

118632 1683

27.3 Interpretation

Funds with HHI values exceeding 1800 demonstrate significant concentration risk and may be more vulnerable to sector-specific downturns.

Business Implication

Investors seeking diversification should carefully evaluate sector exposure before investing.

Figure 13 Placement

Insert:

sector_hhi_chart.png

Caption:

Figure 13: Portfolio Concentration Risk Analysis

28. Fund Recommendation Engine

28.1 Objective

A simple recommendation engine was developed to assist investors in selecting mutual funds based on risk preferences.

The engine categorizes investors into:

- Low Risk
- Moderate Risk
- High Risk

It then identifies the top-performing funds within the corresponding risk category.

28.2 Methodology

Recommendations are based on:

- Sharpe Ratio
- Risk Category
- Overall Fund Quality

The system returns the top three funds matching the investor's selected risk profile.

28.3 Benefits

The recommendation engine provides:

- Faster fund selection
- Personalized recommendations
- Simplified decision-making
- Improved investor experience

29. Power BI Dashboard Summary

The final dashboard consists of four interactive pages.

Industry Overview

Provides insights into:

- AUM Growth
- SIP Growth
- Fund House Performance

Fund Performance

Provides insights into:

- Risk vs Return
- Fund Rankings
- NAV Trends

Investor Analytics

Provides insights into:

- Age Distribution
- Transaction Analysis
- Geographic Participation

SIP & Market Trends

Provides insights into:

- SIP Growth
- Category Inflows
- Benchmark Comparison

The dashboard serves as the primary business intelligence interface for stakeholders.

30. Key Findings

The project generated several important business insights:

1. SIP inflows displayed sustained growth throughout the analysis period.
 2. Investors aged 26–35 contributed the highest transaction volume.
 3. Industry assets remain concentrated among a limited number of major AMCs.
 4. Risk-adjusted performance varied significantly across mutual fund schemes.
 5. Several funds exhibited elevated concentration risk based on HHI analysis.
 6. Earlier investor cohorts contributed substantially more capital than newer cohorts.
 7. Rolling Sharpe analysis revealed changing performance characteristics across market conditions.
 8. Downside risk differed considerably across schemes, emphasizing the importance of VaR and CVaR analysis.
-

31. Recommendations

Based on the analytical findings, the following recommendations are proposed:

For Investors

- Prioritize funds with strong Sharpe and Sortino Ratios.
- Consider downside risk metrics such as VaR and CVaR.
- Diversify investments across multiple categories.
- Monitor concentration risk before investing.

For Fund Houses

- Target younger investors through digital engagement initiatives.
- Expand SIP-focused investment campaigns.
- Improve portfolio diversification where concentration risk is elevated.
- Strengthen investor retention programs.

For Analytics Teams

- Introduce predictive analytics and forecasting models.
- Develop machine-learning-based recommendation systems.

- Integrate real-time NAV and market data.
-

32. Limitations

Although the project achieved its objectives, several limitations remain:

- Historical data was used for analysis.
- Real-time market integration was not implemented.
- The recommendation engine follows rule-based logic.
- External macroeconomic variables were not incorporated.

These limitations provide opportunities for future enhancements.

33. Future Scope

Potential future improvements include:

- Machine Learning Recommendation Systems
- NAV Forecasting Models
- Real-Time Data Integration
- Cloud Deployment
- Automated Reporting
- Mobile Dashboard Applications

These enhancements would further improve analytical depth and user experience.

34. Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully developed an end-to-end analytics platform integrating data engineering, exploratory analysis, performance evaluation, advanced risk modeling, and business intelligence reporting.

The project analyzed more than 46,000 NAV records, 32,778 investor transactions, and 40 mutual fund schemes across multiple datasets. Through the implementation of performance metrics, risk analytics, investor behavior analysis, and interactive dashboards, the solution transformed raw financial data into actionable insights.

The resulting platform enables investors, analysts, and business stakeholders to evaluate fund performance, monitor industry trends, understand investor participation, and make informed decisions based on data-driven evidence.

The project demonstrates the practical application of analytics, financial modeling, and visualization techniques within the mutual fund domain and provides a strong foundation for future enhancements involving predictive analytics and machine learning.

Appendix

Deliverables Submitted

- ✓ ETL Pipeline
- ✓ SQLite Database
- ✓ EDA Notebook
- ✓ Performance Analytics Notebook
- ✓ Advanced Analytics Notebook
- ✓ Power BI Dashboard
- ✓ Dashboard PDF
- ✓ Risk Analytics Reports
- ✓ Recommendation Engine
- ✓ Final Presentation
- ✓ Final Project Report

Technology Stack

Python | Pandas | NumPy | SQLite | SQL | Power BI | Matplotlib | Git | GitHub